CHEAT SHEET UTS · WEEKS 1–8

MK 4 · Operations & Supply Chain Management

Don't mix these up

- Efficiency = low cost; effectiveness = right value. Qualifier = to be considered; winner = to be chosen.
- ERRC: Eliminate & Reduce cut cost; Raise & Create add value.
- Slack 0 = critical path. Crash the cheapest critical activity first. CPI = EV/AC, SPI = EV/PV (<1 = bad).
- Utilisation = used ÷ best operating level. Services target ~70%. Add capacity at the bottleneck.
- Queues come from variability. ρ → 1 makes Lq explode. Constant service halves the queue.

W1 Introduction to Operations & Supply Chain Management

Learning objectives and where this chapter sits · Opening case: efficiency at Southwest Airlines · LO 1–1: what OSCM is — design, operation, improvement · The five supply chain processes (Exhibit 1.3) · Services vs goods: five differences and the service package · The goods–services continuum and product–service bundling · LO 1–2: OSCM careers, the COO and professional societies · LO 1–3: major OSCM concepts over time (Exhibit 1.5) · Sustainability, the triple bottom line, business analytics and current issues · LO 1–4: efficiency, effectiveness, value and productivity · Wall Street efficiency ratios and the Honda example · Wrap-up: five things to carry out of Chapter 1

- OSCM = design, operation and improvement of systems that create and deliver primary products/services; a process transforms inputs into valued outputs.
- Operations transform resources; the supply chain moves information and material to and from operations.
- Sketch the supply network end to end (parka example) before optimising.
- Five supply chain processes: plan, source, make, deliver, return — also in services.
- Five service differences: intangible, interaction, heterogeneous, perishable, package of features.
- Goods–services continuum; bundling (IBM) raises revenue but lowers % margin.
- Careers up to COO; societies: APICS, CSCMP, ISM, PMI.
- Concepts over time: manufacturing strategy, JIT, TQM, BPR, Six Sigma, SCM, e-commerce, analytics; JIT + TQC = lean.
- Sustainability and the triple bottom line; business analytics.
- Efficiency vs effectiveness vs value.
- Five ratios; asset turnover first; compare within industry and pair with an operational metric.

FORMULAS

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Net income ÷ headcount
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Sales ÷ headcount
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Sales ÷ receivables
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COGS ÷ average inventory
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Sales ÷ total assets
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W2 Operations Strategy & Blue Ocean Strategy

Learning objectives and session flow · Strategy starts with sustainability (triple bottom line) · What operations & supply chain strategy is, and how it's formulated · The competitive dimensions: how do customers choose? · Trade-offs: order qualifiers vs order winners · Strategy is delivered by activities: IKEA's activity system · Strategy carries risk — and must be measured (productivity) · Blue Ocean: two kinds of market space and the evidence · Value innovation — the cornerstone · Tools 1–3: strategy canvas, four actions, ERRC grid · Three tests of a good strategy (Southwest) and six value-curve signals · The strategy map and how Blue Ocean completes it · Enel's Innovability: the company, the move and the innovation engine · Four Enel discussion questions (with answer directions) · Wrap-up: five takeaways

- Sustainable strategy = people, profit, planet; much of the impact sits in operations.
- Operations & SC strategy guides resource use for corporate strategy; operations effectiveness implements priorities at minimum cost.
- Formulation: strategic analysis → initiatives → implementation projects.
- Competitive dimensions: cost, speed, reliability, quality, demand changes, flexibility/new-product speed, support.
- Order qualifiers vs winners; winners migrate to qualifiers.
- IKEA's activity system: advantage from fit.
- Supply chain risk: identify, assess, mitigate (GM 1996).
- Productivity = outputs ÷ inputs; partial gains can hide a total decline.
- Red vs blue oceans; 14% of launches → 61% of profit; the strategic move is the unit of analysis.
- Value innovation: lower cost and higher value together.
- Strategy canvas, four actions, ERRC grid.
- Focus, divergence, tagline; six value-curve signals.
- Enel's Innovability: purpose, structure, culture, capital allocation, open innovation.

FORMULAS

```
Productivity = Outputs ÷ Inputs
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```
Eliminate + Reduce → lower cost · Raise + Create → higher value
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"Sustainability is the goal; innovation is the tool." — Francesco Starace, CEO Enel sejak 2014 (sebelumnya di balik sukses Enel Green Power)
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W3 Design of Products and Services

Learning objectives and roadmap · Why product design is an operations problem · The six phases of generic development · Seven process variants and platform products · Designing for the customer: QFD and the House of Quality · DFMA, concurrent engineering, VA/VE and ecodesign · Designing service products: three factors, complexity & divergence · The economics of a new product: NPV and sensitivity · Toolkits: Design Thinking (IDEO) and SCAMPER · Case A (Apple) and Case B (Shell) in numbers · Seven discussion questions (with answer directions) · Wrap-up: six takeaways

- Design locks in most lifetime cost and sets operational capability.
- Firm boundaries: integrated, design in–build out, orchestrated network, pure marketer; core competency.
- Six phases from planning to ramp-up (market-pull).
- Seven variants: technology-push, platform, process-intensive, customized, high-risk, quick-build, complex systems.
- Platform products leverage one architecture.
- QFD and the House of Quality.
- DFMA/concurrent engineering; VA/VE's four questions.
- Ecodesign: life cycle, system, multicriteria.
- Service design: three factors; complexity and divergence.
- NPV and sensitivity; volume is the strongest lever.
- Design Thinking and SCAMPER.
- Apple's principles; Shell's demand-side failure.

FORMULAS

```
Technical score = Σ (weight × strength);
closing energy = 79
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NPV = Σ [arus kas periode t ÷ (1 + r)t]
```

W4 Project Management

Learning objectives and roadmap · Why project management is an operations problem · How projects are organised: pure, functional, matrix · The triple constraint: scope, time, cost — and quality in the middle · WBS, milestones and the KOODESIGN example · The Critical Path Method: ES, EF, LS, LF, slack · Three time estimates (PERT) and completion probability · Compressing a schedule: fast tracking, crashing and other options · Time–cost models and the crashing procedure · Earned Value Management (EVM) · Choosing the delivery approach by uncertainty · The Whoosh case: Jakarta–Bandung high-speed rail · Wrap-up: six things from the Whoosh case

- A project is temporary and unique; four types.
- Pure, functional, matrix structures and their trade-offs.
- Triple constraint with quality in the middle.
- WBS, work packages, milestones.
- CPM: ES, EF, LS, LF, slack, critical path.
- PERT: TE, variance, Z.
- Compression options; crash only critical, cheapest first.
- Crash cost slope; minimum total cost.
- EVM: PV, EV, AC, CV, SV, CPI, SPI, EAC.
- Delivery methods by uncertainty.
- Whoosh lessons.

FORMULAS

```
Slack = LS − ES = LF − EF
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TE = (a + 4m + b) ÷ 6 · σ² = ((b − a) ÷ 6)²
```

```
Cost per period = (crash cost − normal cost)
÷ (normal time − crash time)
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```
CV = EV − AC · CPI = EV ÷ AC | SV = EV − PV
· SPI = EV ÷ PV
```

W5 Strategic Capacity Planning

Learning objectives and roadmap · Why capacity is a strategic problem · Capacity, best operating level and utilisation · Economies and diseconomies of scale · Capacity focus and capacity flexibility · Determining requirements, cushions and chunks · Considerations in changing capacity · Decision trees: Hackers Computer Store · Service capacity is a different problem · The resource-based view and the VRIN test · From Five Forces to dynamic capabilities: sensing, seizing, transforming · Pelindo & Semarang Container Terminal: adding capacity on sinking ground · Debrief: six things from the Pelindo case

- Capacity is a rate over a stated period; three horizons.
- Lead time, irreversibility, asymmetry.
- Best operating level and utilisation (480 ÷ 500 = 96%).
- Economies → diseconomies; transport sets plant size; scale ≠ productivity.
- Focused factory, PWP, flexibility, economies of scope.
- Forecast → convert → project → cushion → chunk → balance.
- Balance, frequency, external sources, decreasing.
- Decision trees: Hackers EVs and 16% NPVs.
- Service capacity and the ~70% target.
- VRIN(O).
- Five Forces → RBV → dynamic capabilities.
- TPKS lessons.

FORMULAS

```
Utilisation = used ÷ best operating level → 480 ÷ 500 = 96%
```

```
Capacity cushion = (capacity − demand) ÷ demand
```

W6 Process Design and Analysis

Learning objectives and roadmap · What a process is, and why the type matters · The customer order decoupling point · Five layouts and the product–process matrix · Break-even: choosing the equipment (Example 7.1) · Manufacturing process flow design · Customer contact and the service-system design matrix · Customer-introduced variability and four accommodation strategies · Service poka-yokes, scripted processes and three service designs · BPMN 2.0: core vocabulary and when to use it · The service blueprint: five layers, three lines · The KAI case — One Second at the Gambir Gate · Key takeaways from the KAI case

- Customer inside the system; volume and standardisation drive process choice.
- Decoupling point: MTS, ATO, MTO, ETO.
- Five layouts.
- Product–process matrix diagonal.
- Break-even at 640 and 2,000 units.
- Flow-design tools.
- Customer contact and the service-system design matrix.
- Customer-introduced variability and four strategies.
- Poka-yokes (Three Ts), Ritz-Carlton, three service designs.
- BPMN 2.0 essentials.
- Service blueprint layers and lines.
- KAI: the slowest stage sets capacity; baseline first.

FORMULAS

```
$200 × D
```

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$80,000 + $75 × D
```

```
$200,000 + $15 × D
```

```
B: 80,000 ÷ 125 = 640 units · A: 120,000 ÷ 60 = 2,000 units
```

```
Ceiling = registration capacity ÷ share registering at the station = 80 ÷ p
```

W7 Waiting Lines and Simulation

Learning objectives and roadmap · Why lines form and can't simply be removed · Variability is the engine of the queue · The anatomy of a queuing system: five structural choices · Jacobs' four waiting line models and their formulas · Example 10.1: Western National Bank drive-thru · Three more worked decisions · Why queues explode at high utilisation, and Bob & Ray · The assumptions, and where they break · Discrete-event simulation (DES) · Agent-based modelling (ABM) and generative ABM · The Manggarai case: the queue at the largest junction · Case close: five takeaways

- Lines form from randomness; the decision is service cost vs waiting cost.
- Variability drives queues; constant service halves them; Kingman approximation.
- Five structural choices.
- The exponential distribution is memoryless.
- Four Jacobs models; simulate beyond them.
- M/M/1 and M/D/1 formulas; Little's Law.
- Western National Bank example.
- Queues explode at high utilisation.
- Bob & Ray: blocking and starving.
- Assumptions and where they break.
- DES vocabulary and value.
- ABM and GABM.
- Manggarai lessons.

FORMULAS

Two-moment (Kingman): $Wq \approx [\rho \div (1 - \rho)] \times [(Ca^2 + Cs^2) \div 2] \times$ service time

$\rho = \lambda/\mu \cdot Lq = \lambda^2/[\mu(\mu - \lambda)] \cdot Ls = \lambda/(\mu - \lambda) \cdot Wq = Lq/\lambda \cdot Pn = (1 - \rho)\rho^n$

$Lq = \lambda^2/[2\mu(\mu - \lambda)] \cdot Ls = Lq + \lambda/\mu$

Lq from tables · $Ls = Lq + \lambda/\mu \cdot Wq = Lq/\lambda$

$X = T/(T + U)$; D, F from finite tables; $L = N(1 - F); H = FNX; J = NF(1 - X)$

Little's Law: $L = \lambda W$

$P(n \le 3) = 1 - \rho^4 = 68.4\%$ at $\rho = 0.75$; for 95%, $\rho \le 0.473 \rightarrow \mu \approx 32$

Unit I: $\lambda = 10$, $\mu = 12$ constant → $Lq = 2.08$, $Wq = 12.5$ min

$Lq = \rho^2 \div (1 - \rho)$: 0.50 at 50%, 1.63 at 70%, 4.82 at 85%, 18.05 at 95%

Agen klasik menjalankan aturan; agen generatif diberi prompt berisi konteks z pada waktu t, f(z), lalu LLM mengembalikan tindakan.

W8 Lean Operations & Supply Chain

Learning objectives and roadmap · Lean glossary (to help read the slides) · Lean starts with value, not cost · The seven wastes · Toyota Production System: five design choices and their preconditions · The Toyota house · Sizing a kanban loop · Inventory hides problems — and what lean assumes · Lean tools: what each buys and what it costs · From lean plant to lean supply chain · Lean in services · The Lean Startup: Build–Measure–Learn · What an MVP is and isn't — and when the Lean Startup fits · Working through the four KAI Logistik questions · Six takeaways

- Lean = removing waste; value = what the customer will pay for (defined outside); waste is seen from inside the process.
- Typically <10% of elapsed time adds value; 90%+ is waiting, moving, reworking, sitting.
- Seven wastes: overproduction (root), inventory, waiting, transport, motion, over-processing, defects.
- TPS = five design choices — pull, kanban, small lots (setup first), heijunka, quality at source — each with a precondition.
- Toyota house: goal on the roof, JIT and Jidoka pillars, stability and culture foundations.
- Kanban: k = D × L × (1 + S) ÷ C, round up; attack L, not k.
- Inventory hides problems; removing buffers without removing variability just moves the problem.
- Lean chain: suppliers, procurement, warehousing, logistics, customers; outside your walls change must be negotiated.
- Lean services attack time, motion and variation; lean removes waste, not reassurance.
- Lean Startup: waste = anything producing no validated learning; Build–Measure–Learn; MVP = least thing that could disprove the idea; persevere or pivot.

FORMULAS

$k = D \times L \times (1 + S) \div C$